Laboratorul I

Exercitiul 1

```
% a.
x = 0:0.2:3;
disp(x);
```

```
         0    0.2000    0.4000    0.6000    0.8000    1.0000    1.2000    1.4000    1.6000    1.8000    2.
```

```
% b.
y = sin(x) .* exp(-x);
disp(y);
```

```
         0    0.1627    0.2610    0.3099    0.3223    0.3096    0.2807    0.2430    0.2018    0.1610    0.
```

```
% c.
A = [x',y'];
disp(A);
```

```
         0         0
    0.2000    0.1627
    0.4000    0.2610
    0.6000    0.3099
    0.8000    0.3223
    1.0000    0.3096
    1.2000    0.2807
    1.4000    0.2430
    1.6000    0.2018
    1.8000    0.1610
    2.0000    0.1231
    2.2000    0.0896
    2.4000    0.0613
    2.6000    0.0383
    2.8000    0.0204
    3.0000    0.0070
```

```
% d.
dimensiune = size(A);
disp(dimensiune);
```

```
    16     2
```

```
% e.
A(5,:) = 0;
disp(A);
```

```
         0         0
    0.2000    0.1627
    0.4000    0.2610
    0.6000    0.3099
         0         0
    1.0000    0.3096
    1.2000    0.2807
    1.4000    0.2430
    1.6000    0.2018
    1.8000    0.1610
    2.0000    0.1231
```

```
    2.2000    0.0896
    2.4000    0.0613
    2.6000    0.0383
    2.8000    0.0204
    3.0000    0.0070
```

Exercitiul 2

```
%a.
A = rand(4);
disp(A);
```

```
    0.6022    0.4624    0.3225    0.1759
    0.3868    0.4243    0.7847    0.7218
    0.9160    0.4609    0.4714    0.4735
    0.0012    0.7702    0.0358    0.1527
```

```
%b.
rang = rank(A);
disp(rang);
```

```
     4
```

```
%c.
deter = det(A);
disp(deter);
```

```
    0.0695
```

```
%d.
A = A';
disp(A);
```

```
    0.6022    0.3868    0.9160    0.0012
    0.4624    0.4243    0.4609    0.7702
    0.3225    0.7847    0.4714    0.0358
    0.1759    0.7218    0.4735    0.1527
```

```
%e.
A = [1 2 3; 4 5 6; 7 8 0];
b = [5; 8; -7];

x = A \ b;
disp(x);
```

```
   -1
    0
    2
```

```
%f.
diagonala = diag(A);
disp(diagonala);
```

```
     1
     5
     0
```

```
%g.
x = [-1; 0; 2];
y = [-2; -1; 1];

A = x * y';
disp(A);
```

```
     2     1    -1
     0     0     0
    -4    -2     2
```

```
B = y * x';
disp(B);
```

```
     2     0    -4
     1     0    -2
    -1     0     2
```

```
x = x * pi;
disp(x);
```

```
   -3.1416
         0
    6.2832
```

```
B_patrat = B^2;
disp(B_patrat);
```

```
     8     0   -16
     4     0    -8
    -4     0     8
```

```
if det(A) ~= 0
    A_Inversa = inv(A);
    disp(A_Inversa)
else
    disp('Matrice singulara')
end
```

```
Matrice singulara
```

```
A_radicali = sqrt(A);
disp(A_radicali);
```

```
   1.4142 + 0.0000i   1.0000 + 0.0000i   0.0000 + 1.0000i
   0.0000 + 0.0000i   0.0000 + 0.0000i   0.0000 + 0.0000i
   0.0000 + 2.0000i   0.0000 + 1.4142i   1.4142 + 0.0000i
```

```
A_superior = A .* ((1:3) > (1:3)');
disp(A_superior);
```

```
     0     1    -1
     0     0     0
     0     0     0
```

```
maxim = -20;
for i = 1:3
```

```
    if maxim < ( sum(A(:,i)) / 3 )
        maxim = sum(A(:,i)) / 3;
    end
    if maxim < ( sum(B(:,i)) / 3 )
        maxim = sum(B(:,i)) / 3;
    end
end
disp(maxim);
```

```
    0.6667
```

```
for i = 1:3
    disp(max(A(i,:)))
    disp(max(A(:,i)))
    disp(max(abs(A(i,:))))
    disp(max(abs(A(:,i))))
end
```

```
     2
     2
     2
     4
     0
     1
     0
     2
     2
     2
     4
     2
```

```
for i = 1:3
    disp(max(B(i,:)))
    disp(max(B(:,i)))
    disp(max(abs(B(i,:))))
    disp(max(abs(B(:,i))))
end
```

```
     2
     2
     4
     2
     1
     0
     2
     0
     2
     2
     2
     4
```

```
urmaA = trace(A);
disp(urmaA);
```

```
     4
```

```
urmaB = trace(B);
```

```
disp(urmaB);
```

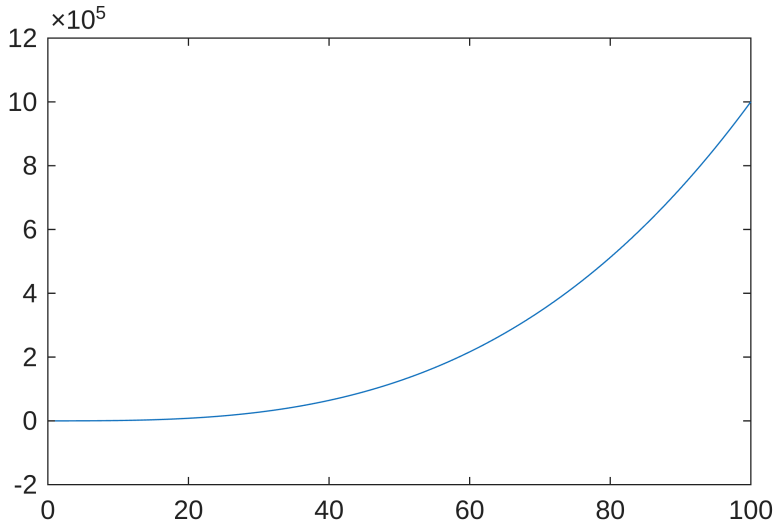
```
     4
```

```
C = [1 1 0 0 0; 1 1 1 0 0; 0 1 1 1 0; 0 0 1 1 1; 0 0 0 1 1];
disp(C);
```

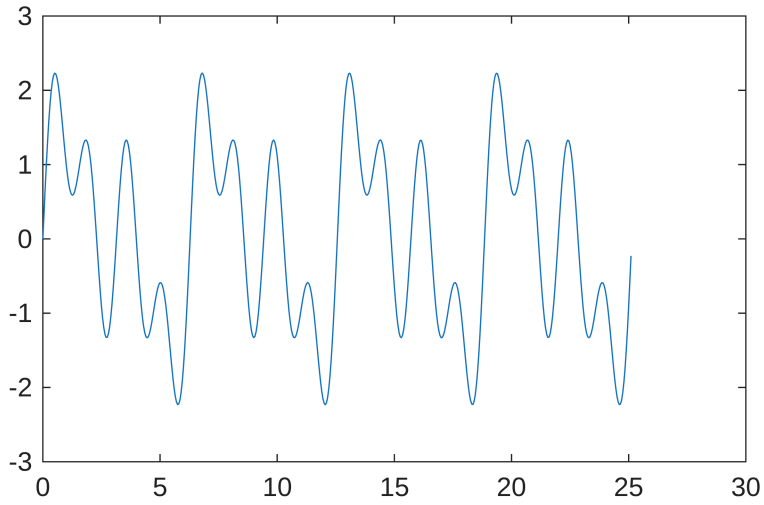
```
     1     1     0     0     0
     1     1     1     0     0
     0     1     1     1     0
     0     0     1     1     1
     0     0     0     1     1
```

Exercitiul 3

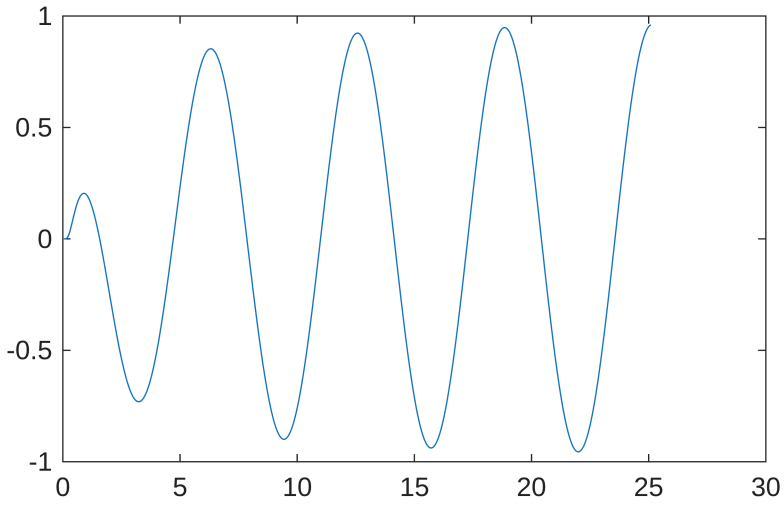
```
% a.
x = 1:100;
y = x.^3 + 4 * x - 15;
plot(x,y)
```



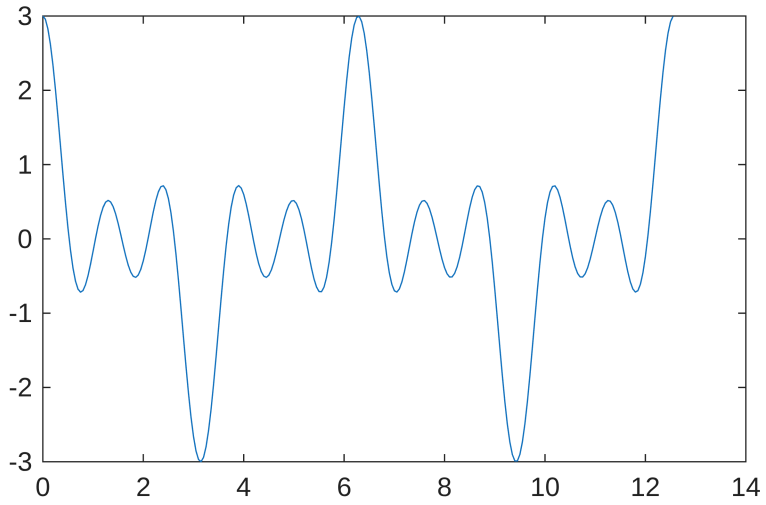

```
% b.
x = 0:0.05:8*pi;
y = sin(x) + sin(2*x) + sin(4*x);
plot(x,y)
```



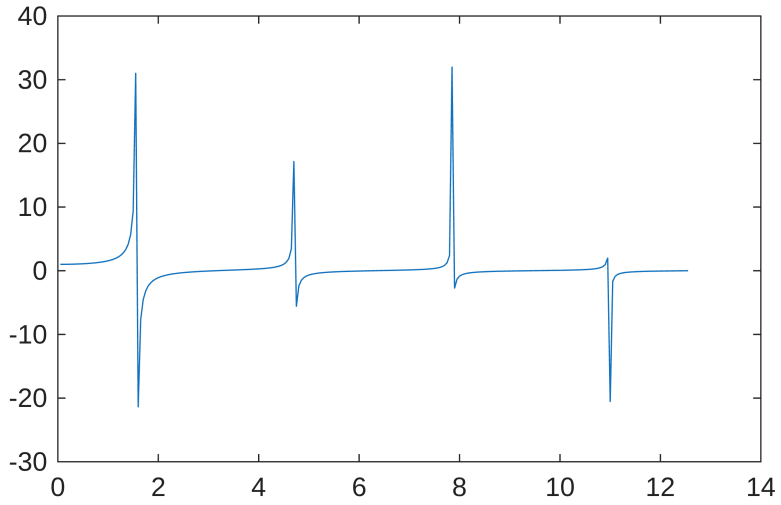

```
% c.
x = 0:0.05:8*pi;
y = exp(-1./x) .* cos(x);
plot(x,y)
```



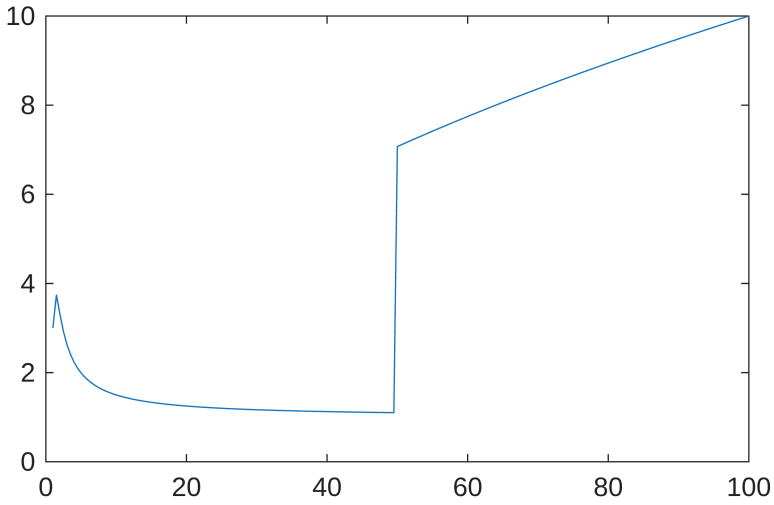

```
% d.
x = 0:0.05:4*pi;
y = cos(x) + cos(3*x) + cos(5*x);
plot(x,y)
```




```
% e.
x = 0:0.05:4*pi;
y = tan(x) ./ x;
plot(x,y)
```




```
% f.
x = 1:0.5:100;
y = (1 + 5./x - 3./(x.^4)) .* (1 <= x & x < 50) + sqrt(x) .* (50 <= x & x <=
100);
plot(x,y)
```


10
8
6
4
2
0
0
20
40
60
80
100